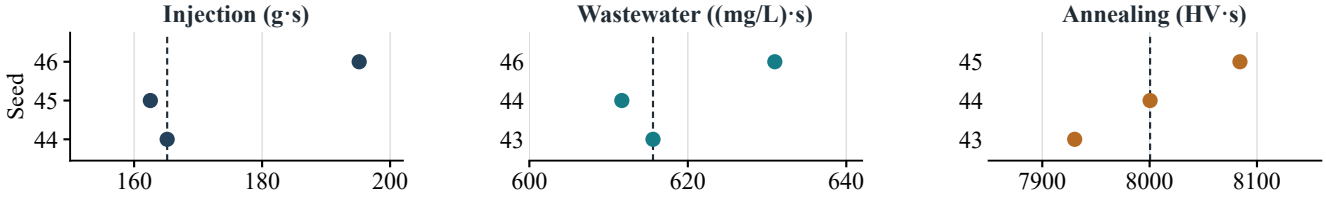
(a) Fixed-horizon process comparison

Delta IAE = IAE(C0) - IAE(C1); positive values favor the script. Horizontal scales are scene-specific.



Points: valid pairs; dashed lines: scene medians. Invalid C0 gaps: wastewater seed 45 (9 s), annealing seed 46 (6 s).

Separate injection seed-43 pilot (ΔIAE = 163.240 g·s) is not plotted.

(b) Process-window guard ablation

Three technical repetitions per arm; benchmark mode.

	Rep. 1		Rep. 2		Rep. 3		
	Below	Above	Below	Above	Below	Above	HTTP 200 / 6
Full guard	○	○	○	○	○	○	0/6
Window check removed	■	■	■	■	■	■	6/6
Readback check removed	○	○	○	○	○	○	0/6
Window + readback removed	■	■	■	■	■	■	6/6

Open circle: HTTP 400 (rejected) | filled square: HTTP 200

No readback-mismatch stimulus was included; HTTP status does not prove a driver call.